English-Chinese Knowledge Conflict Signals Mirror General Cross-Lingual Asymmetries

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Abstract

When multilingual large language models (LLMs) encounter conflicting information across languages, presentation language can determine the answer. Behavioral studies attribute this to linguistic affinity, but the internal mechanism remains unknown: how is the conflict signal encoded, and does detectability imply causal influence on resolution?

We probe residual stream representations in Llama-3.1-8B-Instruct, Qwen3-8B, and Gemma-2-9B-IT, focusing on English (EN)–Chinese (ZH) conflicts. Our analysis combines cross-lingual transfer, Cross-lingual Natural Language Inference (XNLI) non-conflict baselines, confound exclusion, and representation steering. Conflict signals are linearly detectable across languages, and cross-lingual transfer exhibits a directional asymmetry: probes trained on Chinese transfer to English $\sim$7 percentage points more accurately than the reverse. This asymmetry is consistent with the XNLI non-conflict baseline in both magnitude and direction, with overlapping confidence intervals, mirroring the model's general multilingual structure rather than conflict-specific encoding. A dissociation emerges between detection and decision in Llama-3.1-8B-Instruct: the probe signal peaks at late layers (88% depth), but representation steering achieves causal influence only at a mid-layer (41% depth); analogous Qwen and Gemma steering yields no significant effect. These findings, observed in Llama-3.1-8B-Instruct and absent in Qwen and Gemma, indicate that linear detectability does not imply causal relevance, with implications for probe-based intervention.

1 Introduction

When multilingual large language models (LLMs) encounter conflicting information across languages, the language of presentation can determine which answer the model produces. Zhao et al. (2026) showed that linguistic affinity dominates conflict resolution in entity-centric factual conflicts, with models preferring answers from linguistically closer sources. Yet the internal mechanism behind this behavior remains opaque: how does the model encode the conflict signal? Can it be detected from internal representations? And if detectable, does the detected signal causally influence the model's resolution strategy? These questions sit at the intersection of mechanistic interpretability and multilingual NLP, and answering them requires moving beyond behavioral observations to probe model internals directly.

Existing work on probing knowledge conflict signals has been limited to monolingual settings. Zhao et al. (2024) showed that residual stream representations carry linearly detectable conflict signals in English, and Brink et al. (2026) performed knowledge attribution probing to localize factual associations within model parameters. Extending these approaches cross-lingually introduces additional challenges. Translation quality may confound probe transfer: a poor translation could alter conflict detection for reasons unrelated to the conflict signal itself. Token overlap between languages, even when minimal, could provide a shortcut for cross-lingual probes. Observed asymmetries in transfer accuracy may reflect general cross-lingual properties rather than conflict-specific phenomena, and disentangling these sources requires careful baseline comparisons. Beyond detectability, a fundamental limitation of probing is that it establishes correlation, not causation (Hewitt and Liang, 2019). Basu et al. (2026) demonstrated that even near-perfect probe accuracy does not guarantee that the detected direction can steer model behavior.

Our work extends Zhao et al. (2024) in five directions: (1) cross-lingual expansion from monolingual English to EN-ZH conflict probing; (2) a systematic confound exclusion pipeline (translation quality, token overlap, non-conflict baseline);

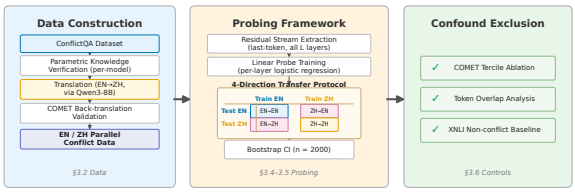


Figure 1: Overview of the cross-lingual conflict probing pipeline. We construct parallel EN/ZH conflict data from ConflictQA with COMET-validated translations, train per-layer logistic regression probes in four transfer directions, and apply three confound exclusion controls (COMET tercile ablation, token overlap analysis, and XNLI non-conflict baseline).

(3) the XNLI baseline revealing that asymmetry mirrors general cross-lingual structure; (4) three-model causal intervention with boundary mapping; and (5) evidence that detectability does not predict causal relevance. We address these questions through a pipeline that combines residual stream probing with representation steering across three models of distinct architectures: Llama-3.1-8B-Instruct, Qwen3-8B, and Gemma-2-9B-IT. Probes achieve 77.6%, 75.0%, and 76.2% balanced accuracy on English within-language conflict detection, respectively, and all cross-lingual transfer directions exceed chance (binomial $p < 0.001$). Cross-lingual transfer exhibits a directional asymmetry: probes trained on Chinese transfer to English approximately 7 percentage points (pp) more accurately than the reverse in all three models. This asymmetry, however, is not conflict-specific. XNLI entailment-versus-contradiction probes (Conneau et al., 2018) show the same directional pattern at comparable magnitude (+6.0, +6.7, and +7.8 pp), with substantially overlapping confidence intervals. The conflict asymmetry thus mirrors a general property of English-dominant models, where Chinese representations project more readily onto English representational space than the reverse. Our causal analysis reveals a dissociation between detection and decision in Llama-3.1-8B-Instruct: the probe signal peaks at late layers (L28, 88% depth), but representation steering succeeds only at a mid-layer (L13, 41% depth; Fisher $p = 0.002$), with all late-layer interventions producing no significant behavioral change. To ensure these findings are robust, we apply a confound exclusion pipeline that includes COMET-based translation quality stratification, token overlap analysis, and non-conflict baseline comparison.

These results carry two methodological implications. The layer dissociation between probing peak and causal influence demonstrates that the layer with highest probe accuracy is not necessarily where the model resolves the conflict. This challenges the common practice of intervening at the probing peak layer, consistent with the warning from Basu et al. (2026) that detectability and causal relevance can diverge. The directional asymmetry in cross-lingual transfer reflects a structural property of English-dominant multilingual models, not a conflict-specific phenomenon. Any cross-lingual probing study must account for this baseline asymmetry to correctly interpret transfer differences.

Our contributions are as follows:

- **Cross-lingual conflict detectability and asymmetry characterization.** We provide the first evidence that knowledge conflict signals are linearly detectable across languages in residual stream representations. The transfer asymmetry (ZH→EN > EN→ZH by ∼7 pp) matches the XNLI non-conflict baseline in both magnitude and direction, establishing that it reflects the model's general cross-lingual structure rather than conflict-specific encoding.

- **Architecture-dependent causal localization.** In Llama-3.1-8B-Instruct, detection accuracy forms a broad plateau across middle-to-late layers, while causal influence is consistently localized to layer 13 (41% depth; Fisher $p = 0.002$). This demonstrates that linear detectability is neither sufficient nor predictive of causal relevance, as confirmed by Gemma's 88.9% probe accuracy with zero causal effect.

- **Confound exclusion pipeline.** We validate a pipeline of COMET translation quality ablation, token overlap analysis, and non-conflict baseline comparison that excludes common confounds in cross-lingual probing.

2 Related Work

Knowledge conflict detection. Xie et al. (2024) introduced ConflictQA and showed that models exhibit context-dependent resolution strategies, and Xu et al. (2024) provided a systematic conflict taxonomy (intra-memory, context-memory, inter-context). Moving beyond behavioral observation, Zhao et al. (2024) demonstrated linearly separable conflict signals in English residual streams,

Brink et al. (2026) localized factual associations via probing, Pham et al. (2026) traced intra-memory conflict pathways, and Li et al. (2025) proposed conflict mitigation methods. These studies operate exclusively in monolingual settings; extending to cross-lingual scenarios introduces confounds (translation quality, token overlap, general transfer properties) that we address with a dedicated exclusion pipeline.

Cross-lingual representations in multilingual models. Intermediate layers develop language-independent representations: Wu et al. (2020) showed cross-lingual structure emerges without explicit alignment, and Muller et al. (2021) found mid-layer alignment in multilingual BERT. For decoder-only LLMs, Wendler et al. (2024) showed that Llama processes non-English inputs through an English-centric latent space, and Kargaran et al. (2024) quantified this bias across language pairs. These findings predict our empirical observations: mid-layer steering success aligns with intermediate-layer language alignment, and the ZH→EN transfer advantage reflects English-centric structure. Behavioral consequences have been documented by Wang et al. (2025), Zhao et al. (2026) (CLEAR cross-lingual conflict benchmark), and Ifergan et al. (2024); Xing et al. (2024); our work provides a mechanistic account of why such asymmetries arise.

Representation steering and causal intervention. Li et al. (2023) introduced Inference-Time Intervention for steering via learned bias vectors, Turner et al. (2023) proposed Activation Addition as an optimization-free alternative, and Zou et al. (2023) generalized these approaches through Representation Engineering. Marks and Tegmark (2024) showed that truth-related representations exhibit linear structure amenable to such interventions. In knowledge conflicts, Zhao et al. (2025) applied Sparse Autoencoder (SAE)-based steering (SpARE) in monolingual settings. Recent work has begun exploring cross-lingual steering: Pokharel et al. (2026) proposed training-free cross-lingual activation steering, and Sundar et al. (2025) analyzed how model interventions induce cross-lingual alignment shifts. These successes are not universal: Basu et al. (2026) showed that near-perfect probe accuracy on clinical NLP tasks failed to translate into effective steering. Feng et al. (2026) proposed fine-grained activation steering targeting specific layers, and Meng et al. (2022) demonstrated localized causal editing via activation patching (Rank-One Model Editing; ROME); alternative causal frameworks (Geiger et al., 2024) remain unexplored for cross-lingual conflict settings. Our results identify layer selection as a critical variable: mid-layer intervention succeeds where late-layer intervention fails, suggesting conflict resolution occurs upstream of where the outcome becomes linearly readable.

3 Method

Our pipeline proceeds from data construction through probing and steering, formalized below.

3.1 Task Formulation

Given a context passage c and question q, the model produces an answer that is either *contextual* (derived from c) or *parametric* (from pretrained knowledge). When c contradicts parametric knowledge, the choice constitutes a knowledge conflict. We train a logistic regression probe $f_\theta(\mathbf{h}_l) = \sigma(\mathbf{w}_l^\top \mathbf{h}_l + b_l)$ on the last-token residual stream representation $\mathbf{h}_l \in \mathbb{R}^d$ at layer l to predict a binary label $y \in \{\text{contextual}, \text{parametric}\}$.

3.2 Data Construction

We build on ConflictQA (Xie et al., 2024), which pairs questions with context passages contradicting the model's parametric knowledge. We filter samples by querying each model without context and retaining only those answered correctly, yielding 1,368 samples for Llama (Grattafiori et al., 2024), 966 for Qwen (Yang et al., 2025), and 1,590 for Gemma (Gemma Team et al., 2024) (80/20 train/test split with stratification). For cross-lingual transfer, we translate all inputs into Chinese using a Qwen3-8B pipeline, with quality validated by COMET (Crosslingual Optimized Metric for Evaluation of Translation; Rei et al. (2020)) back-translation scores used in confound analysis (§3.5).

3.3 Activation Extraction and Probing

For each input, we construct English and Chinese versions and extract the last-token residual stream $\mathbf{h}_l \in \mathbb{R}^d$ at every layer l ($L = 32$ for Llama, 36 for Qwen, 42 for Gemma; $d = 4096$ for Llama/Qwen, 3584 for Gemma), following the last-token aggregation principle (Belinkov, 2022). We record the model's answer and assign binary labels (contextual vs. parametric).

At each layer, we train an L_2-regularized logistic regression probe (Belinkov, 2022; Hewitt and

Liang, 2019) and evaluate with balanced accuracy. We report peak-layer accuracy (maximum across layers) as the primary metric. Confidence intervals are estimated via bootstrap resampling ($n = 2{,}000$) (Ulmer et al., 2022).

3.4 Cross-Lingual Transfer Protocol

We evaluate probes in four directions: EN→EN, ZH→ZH (within-language upper bounds), EN→ZH, and ZH→EN (cross-lingual transfer). We quantify directional asymmetry as $\Delta_{\text{asym}} = \text{acc}(\text{ZH}\rightarrow\text{EN}) - \text{acc}(\text{EN}\rightarrow\text{ZH})$, computed via paired bootstrap (same resampled indices for both directions).

3.5 Confound Exclusion

We exclude two artifacts: (1) *translation quality*, where we stratify test samples by COMET score into terciles and compare probe accuracy (Fisher's exact test); (2) *token overlap*, where we compute per-sample Jaccard similarity between English and Chinese token sets and test for correlation with accuracy (Pearson's r). As a *non-conflict baseline*, we apply the same pipeline to XNLI entailment-versus-contradiction (Conneau et al., 2018); if $\Delta_{\text{asym}} > 0$ also appears there, the asymmetry reflects a general cross-lingual property.

3.6 Representation Steering

To test whether the probe direction is causally relevant (Hewitt and Liang, 2019), we inject $\alpha \cdot \mathbf{w}_{\text{probe}}$ into the last-token residual stream at layer l during inference (Li et al., 2023; Zou et al., 2023): $\mathbf{h}'_l = \mathbf{h}_l + \alpha \cdot \mathbf{w}_{\text{probe}}$, where positive α pushes toward the contextual direction and negative toward the parametric. As control, we inject a random vector of equal norm. We measure the flip rate (fraction of samples switching answer category) and compare via Fisher's exact test. We apply Bonferroni correction per layer ($k = 10$ α values), treating each layer as an independent hypothesis about causal locus, consistent with prior work in mechanistic interpretability that tests layers individually (Meng et al., 2022; Geiger et al., 2024). For Llama L13 (the layer showing causal effects), this yields a corrected threshold of $0.05/10 = 0.005$. For last-token injection, we sweep $\alpha \in \{\pm 20, \pm 10, \pm 5, \pm 2, \pm 1\}$ ($k = 10$) at three Llama layers (L13, L18, L28) spanning 41–88% depth; all-position and cross-model experiments use a reduced 6-value grid ($\alpha \in \{\pm 20, \pm 10, \pm 5\}$). Steering evaluation uses only the conflict-condition subset of the English test data, yielding n_{test} smaller than the full probe test set (e.g., 136 vs. 272 for Llama).

4 Experiments

4.1 Experimental Setup

Models. We probe three instruction-tuned models with comparable parameter counts but distinct architectures: Llama-3.1-8B-Instruct (32 layers, hidden dimension 4096) (Grattafiori et al., 2024), Qwen3-8B (36 layers, hidden dimension 4096) (Yang et al., 2025), and Gemma-2-9B-IT (42 layers, hidden dimension 3584) (Gemma Team et al., 2024).

Dataset. We use ConflictQA (Xie et al., 2024), a benchmark of context-parametric knowledge conflicts where a provided context contradicts the model's parametric knowledge. After verifying that each model can correctly answer the parametric-knowledge question without context, we retain 1,368 samples for Llama (1,096 train / 272 test), 966 for Qwen (773 train / 193 test), and 1,590 for Gemma (1,274 train / 316 test), split 80/20 with stratification. The three models were independently verified, yielding only 3.5% three-way question overlap (23 questions); pairwise overlaps range from 7.2% (Llama ∩ Gemma) to 24.8% (Gemma ∩ Qwen), so the consistent asymmetry pattern across models constitutes largely independent replication.

Translation. We translate all English contexts and questions into Chinese using a Qwen3-8B translation pipeline (EN→ZH), with quality validated by COMET (Rei et al., 2020) back-translation scores.

Probing. For each layer l, we train an L_2-regularized logistic regression probe ($C = 1.0$, scikit-learn default) on the last-token residual stream representation $\mathbf{h}_l$, with binary labels indicating whether the model chose the contextual or parametric answer. We report balanced accuracy to account for class imbalance. All confidence intervals (CIs) are computed via bootstrap resampling with paired differences for asymmetry estimation. All activation extraction and steering experiments were conducted on a single GPU; the boundary replication (Appendix B) used an NVIDIA RTX PRO 6000.

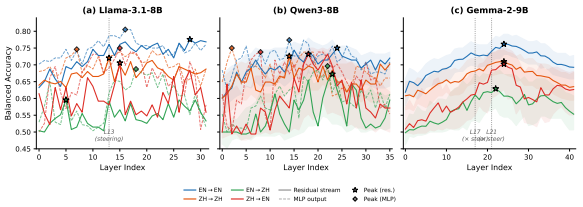


Figure 2: Layer-wise probe balanced accuracy for Llama-3.1-8B (left), Qwen3-8B (center), and Gemma-2-9B-IT (right). Solid lines: residual stream probes; dashed lines (Llama and Qwen only): multilayer perceptron (MLP) sublayer probes. Stars mark peak layers. In the Llama panel, the vertical annotation at L13 marks the layer where steering produces significant causal influence (Fisher $p = 0.002$), coinciding with the ZH→ZH probe peak but not the EN→EN peak (L28). In the Gemma panel, L17 and L21 are annotated as steering-negative layers.

Table 1: Single-seed balanced accuracy (%) at each direction's optimal layer. Δ_{asym} and 95% CIs from paired bootstrap ($n = 2{,}000$ resamples). Bold: best per-task direction.

Direction	Llama-3.1-8B	Qwen3-8B	Gemma-2-9B
Conflict Task			
EN → EN	**77.6**	**75.0**	**76.2**
ZH → ZH	72.1	72.6	71.0
EN → ZH	59.6	67.3	62.9
ZH → EN	70.6	73.2	70.2
Δ_{asym}	+7.3 [0.4, 14.0]	+7.2 [−0.6, 13.7]	+6.6 [1.9, 11.1]
XNLI Baseline			
EN → EN	94.0	96.6	95.6
ZH → ZH	87.5	89.5	89.0
EN → ZH	89.0	90.5	89.4
ZH → EN	**94.9**	**97.2**	**97.5**
Δ_{asym}	+6.0 [0.9, 11.3]	+6.7 [2.4, 11.7]	+7.8 [5.7, 10.0]

Note: Δ_{asym} is the mean of paired bootstrap differences, not the arithmetic difference of the reported single-seed accuracies.

Non-conflict baseline. To test whether cross-lingual transfer patterns are specific to knowledge conflicts, we run the same probing pipeline on XNLI (Conneau et al., 2018), classifying entailment versus contradiction.

4.2 Within-Language Conflict Detection

Table 1 reports balanced accuracy for all probe transfer directions. All three models encode linearly detectable conflict signals in their residual streams. Llama achieves 77.6% on EN→EN (peak at layer 28, 88% depth) and 72.1% on ZH→ZH. Qwen reaches 75.0% on EN→EN (peak at layer 24, 67% depth) and 72.6% on ZH→ZH. Gemma attains 76.2% on EN→EN (peak at layer 24, 57% depth) and 71.0% on ZH→ZH. All within-language accuracies are significantly above chance (binomial $p < 0.001$). Figure 2 shows the layer-wise probe accuracy curves for all three models. These results confirm that knowledge conflict signals are linearly separable in the residual stream across all three architectures and both languages.

4.3 Cross-Lingual Transfer and Asymmetry

All four cross-lingual transfer directions achieve balanced accuracy above chance (binomial $p < 0.001$; Table 1), confirming that conflict representations partially overlap across languages. All three models show a consistent directional trend: probes trained on Chinese representations transfer to English (ZH→EN) more accurately than the reverse direction (EN→ZH). We define $\Delta_{\text{asym}} = \text{acc}(\text{ZH}\to\text{EN}) - \text{acc}(\text{EN}\to\text{ZH})$, computed via paired bootstrap. For Llama, $\Delta_{\text{asym}} = +7.3$ percentage points (pp) with a 95% bootstrap CI of [0.4, 14.0] ($p = 0.038$); for Qwen, $\Delta_{\text{asym}} = +7.2$ pp with CI [−0.6, 13.7]; for Gemma, $\Delta_{\text{asym}} = +6.6$ pp with CI [1.9, 11.1] ($p = 0.006$). While the Qwen CI marginally includes zero, the asymmetry is consistent in direction across all three architectures. The magnitude of this asymmetry is consistent in direction and magnitude with the XNLI non-conflict baseline across all three architectures (§4.4), and the consistent direction across all three models suggests a shared structural property rather than a model-specific artifact (Figure 3).

4.4 Non-Conflict Baseline Comparison

If the ZH→EN transfer advantage were specific to knowledge conflicts, it should be absent or substantially reduced in a non-conflict semantic task. We test this prediction using XNLI entailment-versus-contradiction probes. All three models achieve high within-language accuracy on XNLI (Llama 94.0% EN, 87.5% ZH; Qwen 96.6% EN, 89.5% ZH; Gemma 95.6% EN, 89.0% ZH), confirming that the probing pipeline captures semantic distinctions beyond knowledge conflicts. The XNLI probes exhibit the same directional asymmetry: Llama $\Delta_{\text{asym}} = +6.0$ pp, CI [0.9, 11.3]; Qwen $\Delta_{\text{asym}} = +6.7$ pp, CI [2.4, 11.7]; Gemma $\Delta_{\text{asym}} = +7.8$ pp, CI [5.7, 10.0] ($p < 0.001$). All three XNLI CIs exclude zero and overlap substantially with the corresponding conflict-task CIs (Figure 3). Formal Two One-Sided Tests (TOST) equivalence tests (±2 pp margin) were non-significant (all $p > 0.36$); post-hoc power analysis shows only 1.4% power at this margin given standard error (SE)

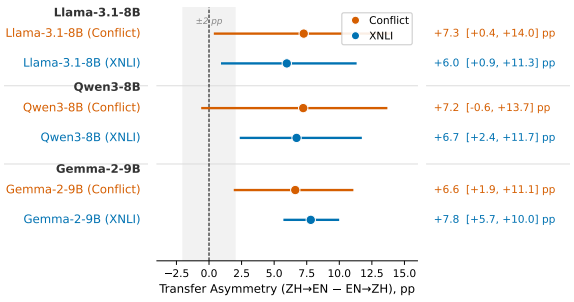


Figure 3: Transfer asymmetry (Δ_{asym}) with bootstrap 95% CIs ($n = 2{,}000$) for conflict and XNLI tasks across all three architectures (six data points). Conflict and XNLI CIs overlap substantially within each model, indicating that the ZH→EN advantage is a general cross-lingual property rather than conflict-specific. Grey band: ±2 pp equivalence zone.

≈ 4.2 pp, with 80% power requiring a ± 12.5 pp margin. Bayesian equivalence testing yields a Bayes factor $\text{BF}_{01} \approx 1.5$ (half-normal prior, scale $= 5$ pp), providing directionally consistent but modest evidence for equivalence. Combined with the consistent direction across all six model×task combinations (binomial $p = 0.016$) and magnitudes clustering in [+6.0, +7.8 pp], these analyses support the "mirror" characterization: the non-significance reflects insufficient power, not meaningful differences.

4.5 Confound Exclusion

Two potential confounds could inflate cross-lingual probe accuracy: translation quality and lexical overlap between English and Chinese inputs.

Translation quality. We stratify test samples by COMET score into terciles (high, medium, low) and compare EN→ZH probe accuracy across strata. If translation quality drove probe transfer, high-COMET samples would yield higher accuracy. Table 2 shows no such pattern: Qwen shows a reversed association ($\Delta = -5.6$ pp, Fisher $p = 0.695$) and Llama a negligible one ($\Delta = +1.7$ pp, $p = 0.761$), both ruling out translation quality as a confound. Gemma shows a non-significant trend in the confound direction ($\Delta = +10.3$ pp, $p = 0.278$), but interpretation is limited by the near-chance baseline accuracy of the EN→ZH probe at L17 (58.8%), which places this analysis in a low-power regime. The minimum detectable difference at 80% power is 25.3 pp ($n \approx 56$ per tercile), far exceeding observed effects.

Table 2: Confound exclusion analysis. COMET tercile: Δ = balanced accuracy difference between high and low COMET score terciles for EN→ZH probes (Fisher's exact test). Token overlap: Pearson correlation between per-sample Jaccard similarity and EN→ZH probe accuracy.

Confound	Model	Result	p
COMET tercile	Qwen	$\Delta = -5.6$ pp	0.695
COMET tercile	Llama	$\Delta = +1.7$ pp	0.761
COMET tercile	Gemma	$\Delta = +10.3$ pp	0.278
Token overlap	Qwen	n.s.	>0.4
Token overlap	Llama	$r = -0.15$	0.012

Token overlap. Per-sample Jaccard similarity between English and Chinese tokenized inputs averages 3–4%, consistent with the minimal lexical overlap between the two languages. Neither model shows a positive correlation between token overlap and probe accuracy: Qwen yields no significant correlation ($p > 0.4$), and Llama shows a weak negative correlation ($r = -0.15$, $p = 0.012$, $r^2 = 0.023$), where higher overlap is associated with *lower* accuracy, running counter to the confound hypothesis. As additional evidence, we translate the same Llama ConflictQA samples into German using the identical Qwen3-8B pipeline and extend the token overlap analysis to EN-DE, where per-sample Jaccard similarity (14.8%) is approximately 4× higher than EN-ZH (3.8%). Despite this substantially higher lexical overlap, the EN↔DE transfer asymmetry is near-zero ($\Delta = +1.2$ pp, n.s.), further ruling out shared tokens as a driver of the observed EN-ZH asymmetry.

4.6 Causal Evidence via Representation Steering

The probing results establish that conflict signals are detectable, but detection does not imply causal relevance. We test whether the probe weight direction $\mathbf{w}_{\text{probe}}$ causally influences conflict resolution by injecting $\alpha \cdot \mathbf{w}_{\text{probe}}$ into the last-token residual stream at a target layer during inference. We measure the flip rate (fraction of samples that switch answer category) and compare against a random-direction control of equal magnitude via Fisher's exact test.

L13 (mid-layer, 41% depth). Steering at layer 13 produces bidirectional causal effects (Table 3). Positive steering ($\alpha = +5$) yields a flip rate of

19.9% versus 6.6% for the random control (Fisher $p = 0.002$; 4 p→c flips), while negative steering ($\alpha = -20$) yields 36.0% versus 17.6% ($p < 0.001$; 5 c→p flips); both survive Bonferroni correction. All-position injection at L13 produces identical results, confirming that the causal signal is concentrated at the final token. Steering at nearby L16 (50% depth, just three layers away) showed no significant effect (all $p > 0.12$), indicating a narrow causal window. Under a conservative family-wise correction across all 30 tests (3 layers $\times$ 10 α values, threshold $p < 0.00167$), L13 remains significant ($\alpha = +10$, $p = 0.0003$). An independent replication ($N = 201$, six layers L12–L28) confirmed L13 significance ($p < 0.001$) and localized the causal boundary: effects attenuate from L14 ($p = 0.002$) through L16 ($p = 0.005$) to non-significance at L18 ($p = 0.010$, n.s. after Bonferroni) and L28 ($p = 0.284$; Appendix Table 4).

Negative layers and coherence collapse. Steering at L28 (probe peak, 88% depth) produces no significant effect with either last-token or all-position injection (all $p > 0.3$); L18 (MLP peak, 56%) is similarly negative ($p > 0.44$). At extreme magnitude ($\alpha = +20$), the control flip rate exceeds steering ($p = 0.053$), indicating coherence collapse.

Causal localization (Llama). In Llama-3.1-8B-Instruct, causal influence is consistently localized to L13 (41% depth) regardless of where detection accuracy peaks (EN→EN at L28, ZH→ZH at L13): late layers encode the outcome in a linearly readable form, while mid-layers host the computation that determines it. An independent replication with a separately trained probe (seed=123, different train/test split) confirms the L13 causal effect: 4/6 α configurations survive Bonferroni correction (vs. 2/6 in the original; Appendix C).

Cross-model steering. Qwen steering at L14 (39%, $n_{\text{test}} = 131$; probe accuracy 71.0%) produced no significant effect at any α (best $p = 0.785$) and zero directional flips. Gemma-2-9B-IT L17 (40% depth) provides a stronger test: with 88.9% probe accuracy, the highest among all three models (exceeding both Llama 76.1% and Qwen 71.0%), steering produced no significant change at any α (best $p = 0.448$) and zero directional flips ($n_{\text{test}} = 171$), ruling out probe quality as a confound and confirming that the causal influence of probe-weight directions is architecture-dependent.

Table 3: Representation steering results. Flip rate: fraction of test samples switching answer category. Control: random-direction perturbation of equal magnitude. Bonferroni-corrected significance threshold $p < 0.005$ ($k = 10$). Significant results in bold.[†]

Layer	Depth	α	Steer (%)	Ctrl (%)	Fisher p	Dir.
L13	**41%**	**+5**	**19.9**	**6.6**	**0.002**	**4 p→c**
L13	**41%**	**−20**	**36.0**	**17.6**	**< 0.001**	**5 c→p**
L16	50%	+5	8.8	8.1	1.000	
L16	50%	+10	18.4	11.0	0.123	
L16	50%	+20	22.8	24.3	0.886	
L18	56%	+2	3.7	1.5	0.447	
L28	88%	−20	14.7	16.9	0.740	
L28 (all)	88%	−20	11.8	11.8	1.000	
Qwen3-8B ($n_{test} = 131$)						
L14	39%	best	n.s.		0.785	0 dir.
Gemma-2-9B-IT ($n_{test} = 171$)[‡]						
L17	40%	best	n.s.		0.448	0 dir.

[†] At $\alpha = +20$, the control flip rate exceeds steering ($p = 0.053$), indicating coherence collapse at extreme perturbation magnitudes. [‡] Gemma: re-run after correcting a label assignment error in an earlier version of the data preparation pipeline.

Appendix D reports matched-condition results at three layers per model (∼36–43% depth), all confirming the null. Cross-task language probe transfer further validates this shared subspace (§5).

5 Discussion

Causal influence is localized to L13 regardless of detection peak. In Llama, causal influence is consistently localized to layer 13 (41% depth; Fisher $p = 0.002$), regardless of where detection accuracy peaks: EN→EN detection peaks at L28 (88% depth) while ZH→ZH peaks at L13 itself, yet causal influence is found only at L13 in both cases, with all late-layer interventions producing no significant behavioral change ($p > 0.3$). This layer coincides with the site of cross-lingual alignment identified in prior work (Muller et al., 2021; Wendler et al., 2024), suggesting that mid-layers serve a dual role in alignment and conflict resolution. The detection signal at later layers is best understood as a diagnostic readout of a decision already made upstream. A practical implication follows: intervention experiments that target only the highest-accuracy layer risk missing the computationally relevant site; cross-layer sweeps are necessary. Steering failed in both Qwen (L14, 39%; $n = 131$, best $p = 0.785$) and Gemma (L17, 40%; $n = 171$, best $p = 0.448$), with zero directional flips in both cases. The Gemma result is particularly informative: its 88.9% probe accuracy, the highest among all three models, rules out probe

quality as a confound, since the direction that best separates conflict labels carries no causal influence despite being highly discriminative. This establishes that the causal influence of the probe direction is architecture-dependent, not a general property of mid-layer computation.

Why does steering fail in Qwen and Gemma? Three mechanistic hypotheses merit consideration. First, the causal pathway may be non-linear: the probe direction captures the dominant linear component of the conflict signal, but the actual causal structure may involve non-linear interactions that single-direction perturbation cannot address (Elhage et al., 2022). Second, residual stream superposition may play a role: Gemma's 88.9% probe accuracy indicates the signal exists, but it may be superposed with other features such that perturbing one direction simultaneously disrupts unrelated computations (Templeton et al., 2024). Third, architectural differences (e.g., Gemma's sandwich normalization) may alter how residual stream perturbations propagate through subsequent layers.

Conflict signals are encoded in a general cross-lingual space. The ZH→EN transfer advantage is consistent across all three architectures (+7.3, +7.2, and +6.6 pp for Llama, Qwen, and Gemma respectively), and the XNLI baseline comparison (+6.0, +6.7, and +7.8 pp) is consistent with the conflict task asymmetry in magnitude across all three models. The asymmetry is therefore not conflict-specific; conflict detection piggybacks on the model's pre-existing cross-lingual alignment structure. This interpretation connects to the linguistic affinity effects reported by Zhao et al. (2026): the CLEAR benchmark found that conflict resolution preferences vary by language pair, a pattern that may mirror the quality of underlying representational alignment documented by Kargaran et al. (2024) and Wendler et al. (2024). Any cross-lingual probing study must account for this baseline asymmetry before attributing transfer differences to task-specific phenomena.

Direct evidence for a shared cross-lingual subspace. A language identity probe (EN vs. ZH) trained on conflict task activations and evaluated on XNLI activations achieves ≥ 0.96 accuracy at most layers across all three models (≥ 0.98 in the reverse direction), confirming that both tasks encode language identity in a common representational subspace. At Llama L13, conflict→XNLI transfer is 0.998 [0.993, 1.000]. Gemma shows reduced mid-layer transfer at L21 (0.730 [0.653, 0.828]), but L17 remains near-perfect (0.994) despite steering failure, suggesting that cross-lingual subspace disruption is not the sole factor behind negative steering outcomes.

Typological distance and asymmetry magnitude. The near-zero asymmetry for EN-DE ($\Delta = +1.2$ pp, CI $[-4.8, +8.1]$, n.s.) contrasts with the consistent +6–8 pp for EN-ZH, consistent with asymmetry magnitude reflecting representational alignment distance. With only two language pairs, we report this as an observation. EN-DE token overlap (Jaccard 14.8%) is $\sim 4\times$ higher than EN-ZH (3.8%), yet the transfer is near-symmetric, ruling out shared tokens as a driver.

Relation to interpretability without actionability. Basu et al. (2026) reported that steering failed in clinical NLP tasks despite near-perfect probe accuracy (98.2% area under the receiver operating characteristic (AUROC)). Our results present a partial contrast: mid-layer steering succeeds with a weaker probe signal, but the key difference is layer selection. Basu et al. steered at layers selected for probe accuracy; our L28 (probe peak) also fails, with causal influence found only at L13. Task characteristics may also contribute: knowledge conflict resolution involves a discrete choice between two competing sources, which may concentrate causal structure at a single computational stage.

A suggestive MLP trend. MLP sublayer probes for EN→ZH show positive conflict-task gains (Qwen +3.84 pp, Llama +5.21 pp) while the XNLI baseline yields negative or zero gains, but both confidence intervals include zero and this pattern requires larger-scale verification.

6 Conclusion

We showed that knowledge conflict signals in multilingual LLMs are linearly detectable across languages (balanced accuracies 75.0–77.6% within-language, all cross-lingual transfers above chance at $p < 0.001$) and that the transfer asymmetry (ZH→EN exceeding EN→ZH by ∼6–7 pp) is consistent with the XNLI non-conflict baseline across all three architectures, reflecting general cross-lingual structure rather than conflict-specific encoding.

In Llama-3.1-8B-Instruct, causal influence is consistently localized to L13 (41% depth; Fisher

$p = 0.002$) regardless of where detection accuracy peaks, and all late-layer interventions fail. Linear detectability is neither sufficient nor predictive of causal relevance; cross-layer sweeps are necessary to identify the computationally relevant site.

Steering failed in both Qwen and Gemma despite Gemma's 88.9% probe accuracy, confirming that the causal influence of the probe direction is architecture-specific. Future work should explore alternative intervention methods, diverse language pairs, and sparse autoencoder decomposition.

Limitations

Our study has three main limitations. First, we evaluate only one language pair (English-Chinese) and causal steering succeeded only in Llama at L13; Gemma L17 failed despite 88.9% probe accuracy ($n = 171$) and Qwen L14 ($n = 131$) was also negative. Typologically diverse language pairs and alternative intervention methods are needed to assess generality. Second, the effective dataset size (966 to 1,590 samples per model) is constrained by parametric knowledge verification, which retains only questions where the model confidently knows the ground truth and may bias toward well-represented knowledge; additionally, Qwen3-8B serves as both the translation model and an experimental subject, though COMET-based quality control and Llama/Gemma cross-validation mitigate this dual-role concern. Third, all probing results are based on a single train/test split with bootstrap confidence intervals capturing within-split variability; while peak balanced accuracy remains stable across seeds (Llama $\pm 0.12\%$, Gemma $\pm 1.32\%$, Qwen $\pm 1.90\%$), the specific peak layer can shift, and future work should verify stability across larger datasets to improve statistical power for stratified comparisons such as the COMET tercile ablation.

A Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform related-work section.

B Fine-Grained Causal Boundary Analysis

Table 4 reports steering results across the full layer gradient from an independent replication ($N = 201$, Llama-3.1-8B-Instruct, RTX PRO 6000).

Table 4: Causal boundary steering results (independently sampled, $N = 201$, Llama-3.1-8B-Instruct). Bonf. Sig. αs: number of α values reaching significance after Bonferroni correction out of 6 tested.

Layer	Depth	Probe Acc	Best α	Best Fisher p	Bonf. Sig. αs	Result
L12	38%	91.75%	+20	0.085	0/6	—
L13	**41%**	**90.00%**	**+10**	**< 0.001**	**2/6**	✓✓
L14	44%	91.75%	+10	0.002	1/6	✓
L16	50%	91.50%	+10	0.005	1/6	✓
L18	56%	90.75%	+5	0.010	0/6	—
L28	88%	85.25%	+10	0.284	0/6	—

C Seed Replication of L13 Steering

To verify that the L13 causal effect is not an artifact of a specific train/test split, we repeated the Llama L13 steering experiment with an independently trained probe (seed=123, different train/test split). The causal effect fully replicates: $\alpha = +5$ ($p = 0.002$), $\alpha = +10$ ($p = 0.0005$), $\alpha = -10$ ($p = 0.0003$), $\alpha = +20$ ($p = 0.004$), with 4 of 6 configurations surviving Bonferroni correction (vs. 2/6 in the original). The consistent positive-direction effect across independent probe training confirms that L13 causal localization is not an artifact of a specific train/test split.

D Matched-Condition Steering Results

Table 5 reports steering results under matched conditions: all three architectures are tested at comparable relative depths (∼36–43%) using the same probe training methodology and six α values in $[-20, +20]$. Despite the highest probe accuracy among the three models (88–89%), Gemma produces zero directional flips, ruling out probe quality as a confound for the Llama-specific positive result. The full Llama causal gradient ($N = 201$) is in Appendix B.

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Table 5: Matched-condition steering results. Probe: EN→EN balanced accuracy at the target layer. Best p: minimum Fisher exact p-value across six tested α values. Bonf.: α values surviving Bonferroni correction out of 6. Flips: total directional transitions (p→c + c→p).†

Layer	Depth	Probe (%)	Best p	Bonf.	Flips
Llama-3.1-8B-Instruct ($N = 136$; 32 layers)					
L13	**41%**	**76.1**	**< 0.001**	**2/6**	**9**
Qwen3-8B ($N = 131$; 36 layers)					
L13	36%	71.8	0.168	0/6	0
L14	39%	71.0	0.785	0/6	0
L15	42%	72.9	0.617	0/6	0
Gemma-2-9B-IT ($N = 171$; 42 layers)†					
L16	38%	88.0	0.067	0/6	0
L17	40%	88.9	0.448	0/6	0
L18	43%	89.5	0.219	0/6	0

† Gemma: re-run after correcting a label assignment error in an earlier version of the data preparation pipeline.

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